Database Architecture — CredVault

System: CredVault Credit Card Management Platform
Version: 1.0
Date: April 2026

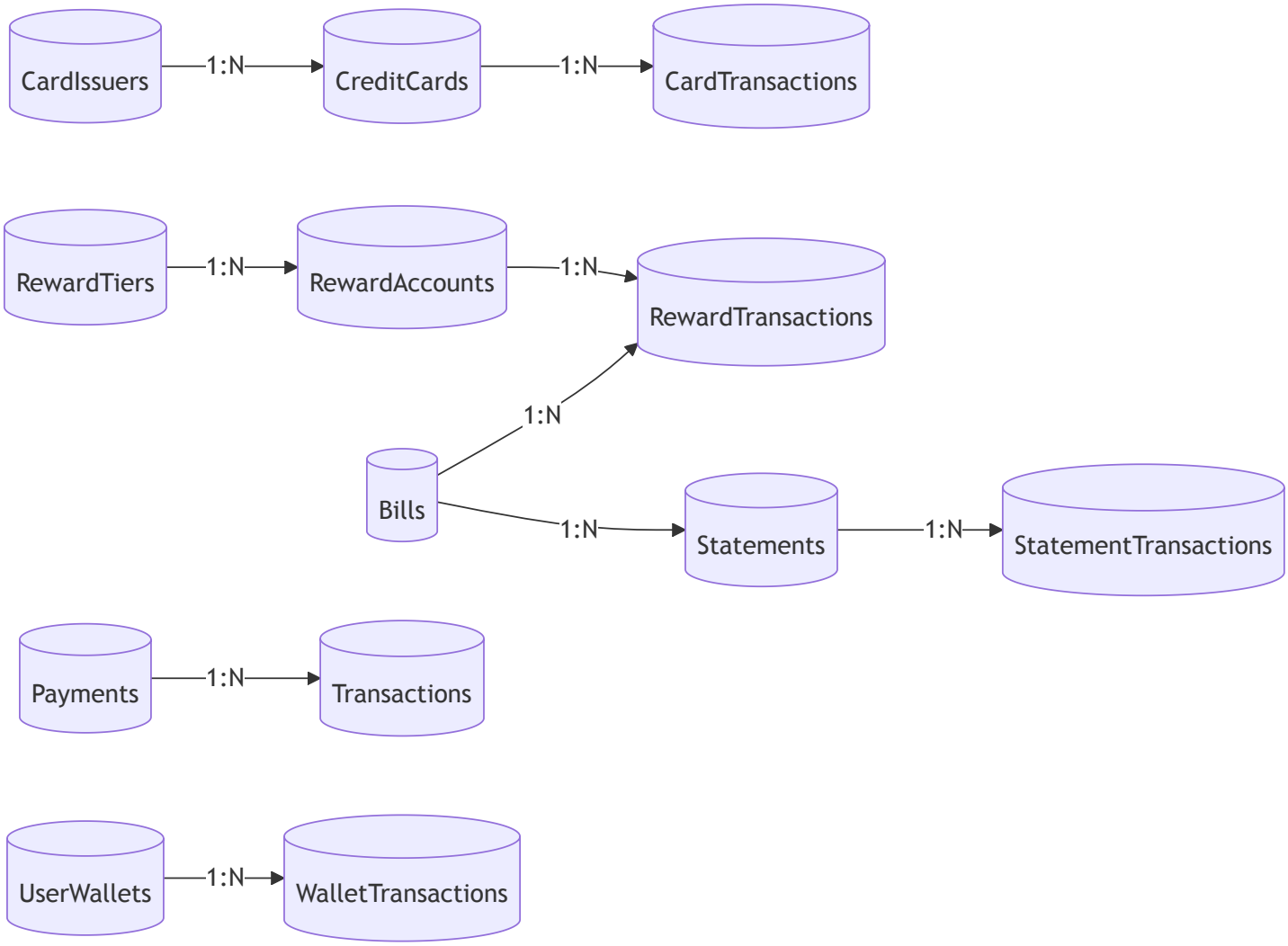
1. Overview

CredVault follows the **database-per-service** pattern. Each of the 5 microservices owns its own isolated SQL Server database. There are **no cross-service foreign key constraints** — referential integrity across services is maintained through event-driven communication and saga orchestration.

Service	Database	Tables	ORM
Identity	`credvault_identity`	1	EF Core (Code-First)
Card	`credvault_cards`	3	EF Core (Code-First)
Billing	`credvault_billing`	6	EF Core (Code-First)
Payment	`credvault_payments`	6	EF Core (Code-First)
Notification	`credvault_notifications`	2	EF Core (Code-First)
Total: 5 databases, 18 tables, 0 cross-service FK constraints.			

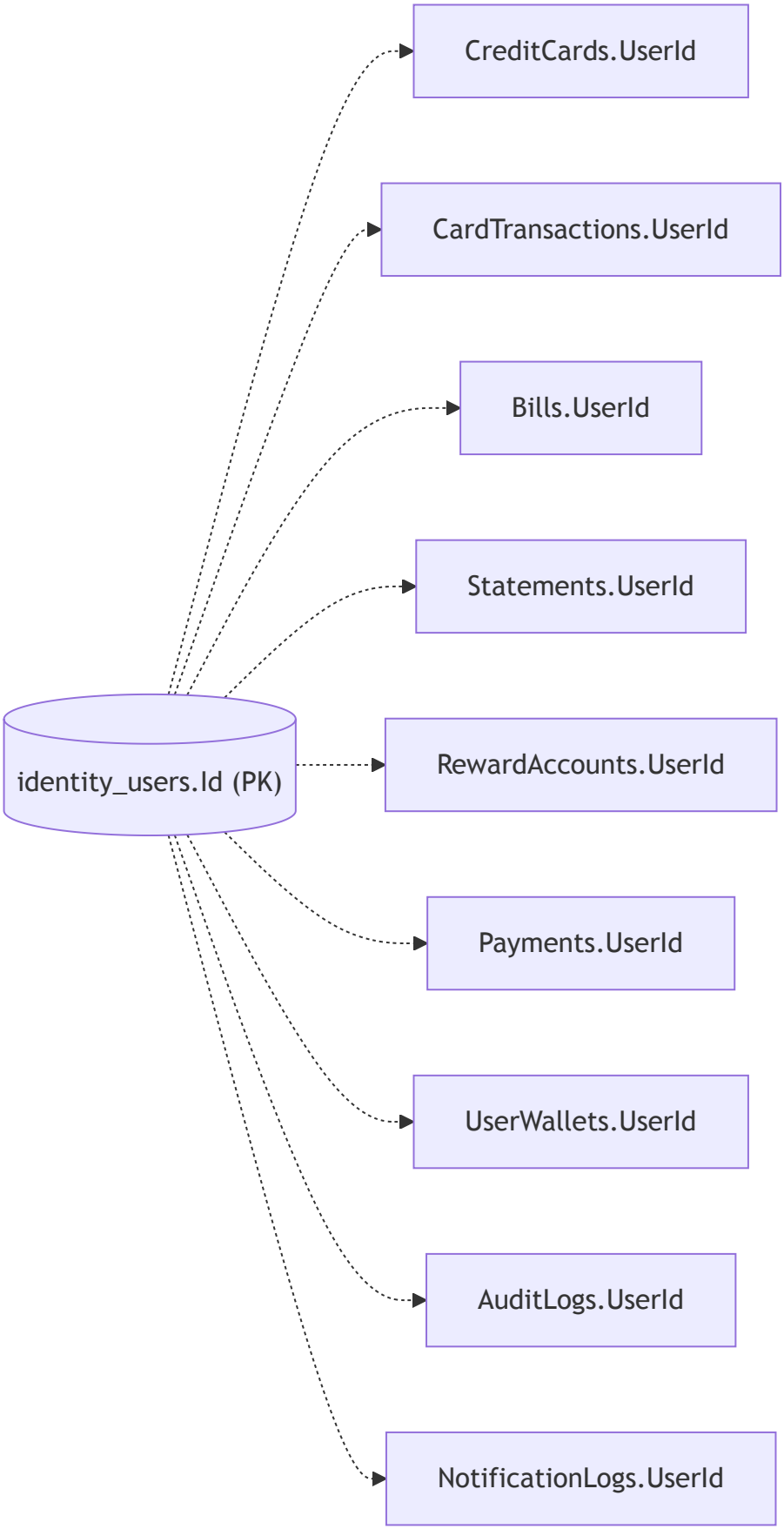
2. Cross-Service Relationship Architecture

2.1 The Core Principle



Legend: Solid arrows = Physical FKs (within same DB). No arrows between services = No cross-service FKs.

2.2 Logical References (GUIDs, No FK Constraints)



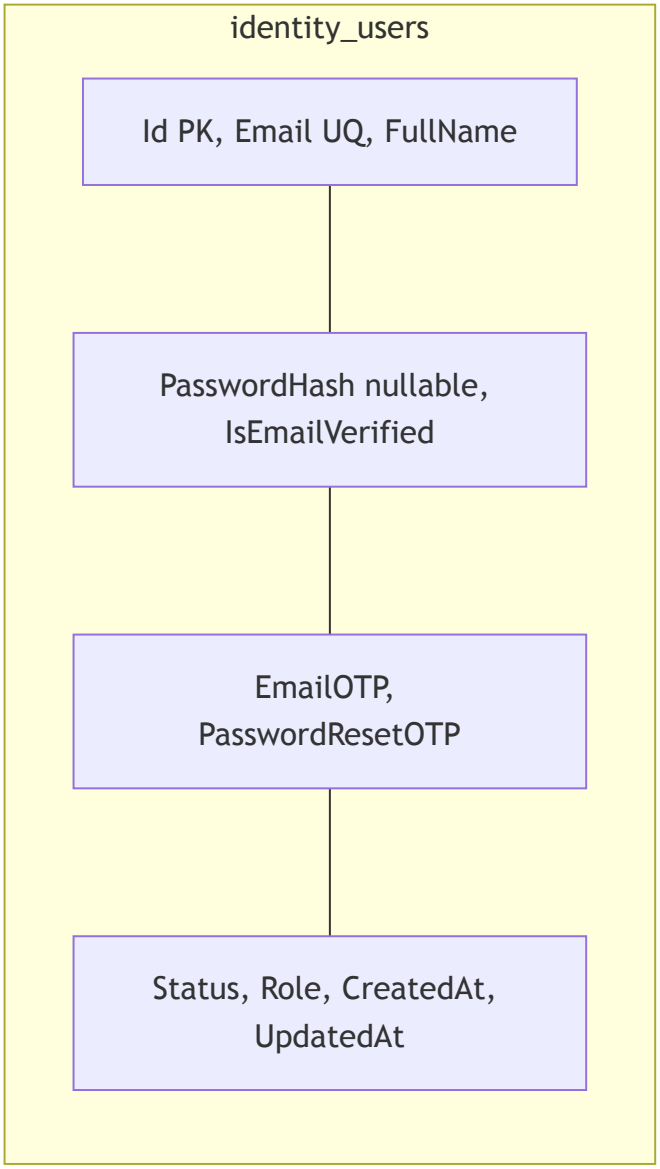
The `identity_users.Id` is referenced by 9 columns across 4 databases — none enforced as foreign keys.

2.3 Complete Cross-Service Reference Map

Source Table	Column	Target Table	Consistency
CreditCards	UserId	identity_users	Event-driven
CardTransactions	UserId	identity_users	Event-driven
CardTransactions	CardId	CreditCards	**Physical FK**
Bills	UserId	identity_users	Event-driven
Bills	CardId	CreditCards	Event-driven
Bills	IssuerId	CardIssuers	Event-driven
Statements	UserId	identity_users	Event-driven
Statements	CardId	CreditCards	Event-driven
Statements	BillId	Bills	**Physical FK**
RewardAccounts	UserId	identity_users	Event-driven
RewardAccounts	RewardTierId	RewardTiers	**Physical FK**
RewardTransactions	RewardAccountId	RewardAccounts	**Physical FK**
RewardTransactions	BillId	Bills	**Physical FK**
Payments	UserId	identity_users	Event-driven + Saga
Payments	CardId	CreditCards	Event-driven + Saga
Payments	BillId	Bills	Event-driven + Saga
Transactions	PaymentId	Payments	**Physical FK**
UserWallets	UserId	identity_users	Event-driven
WalletTransactions	WalletId	UserWallets	**Physical FK**
AuditLogs	UserId	identity_users	Logging only
NotificationLogs	UserId	identity_users	Logging only

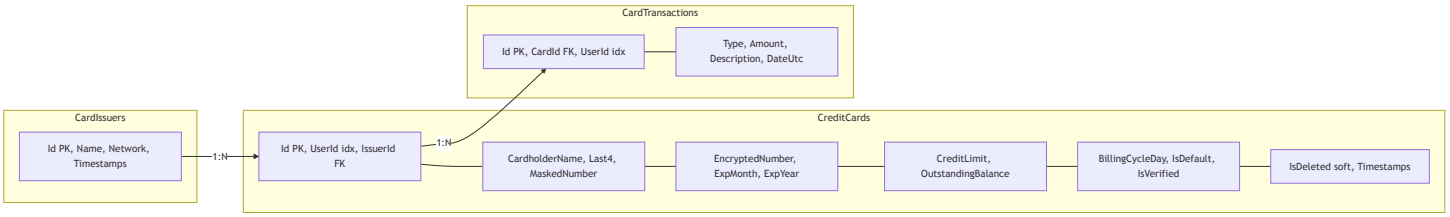
3. Per-Service Database Details

3.1 credvault_identity (1 table)



- **PK:** Id | **UQ:** Email
- `PasswordHash` nullable for Google SSO users
- OTPs are 6-digit, regenerated on each request

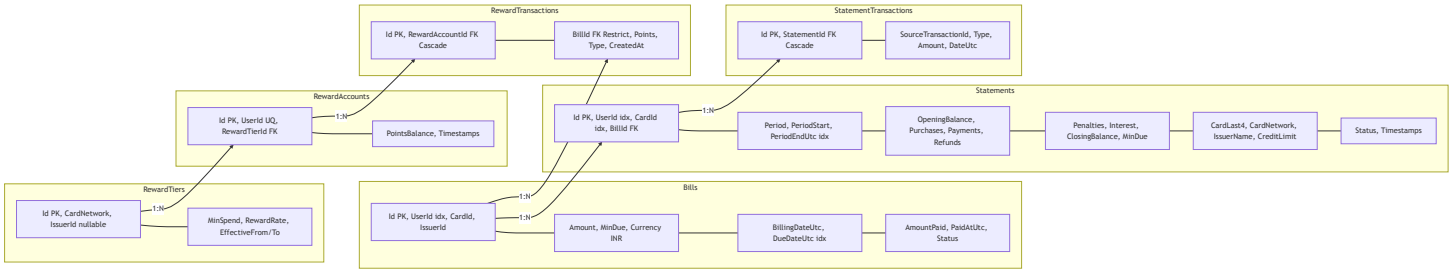
3.2 credvault_cards (3 tables)



Physical FKs: `CreditCards.IssuerId` → `CardIssuers.Id` (Restrict), `CardTransactions.CardId` → `CreditCards.Id` (Restrict)

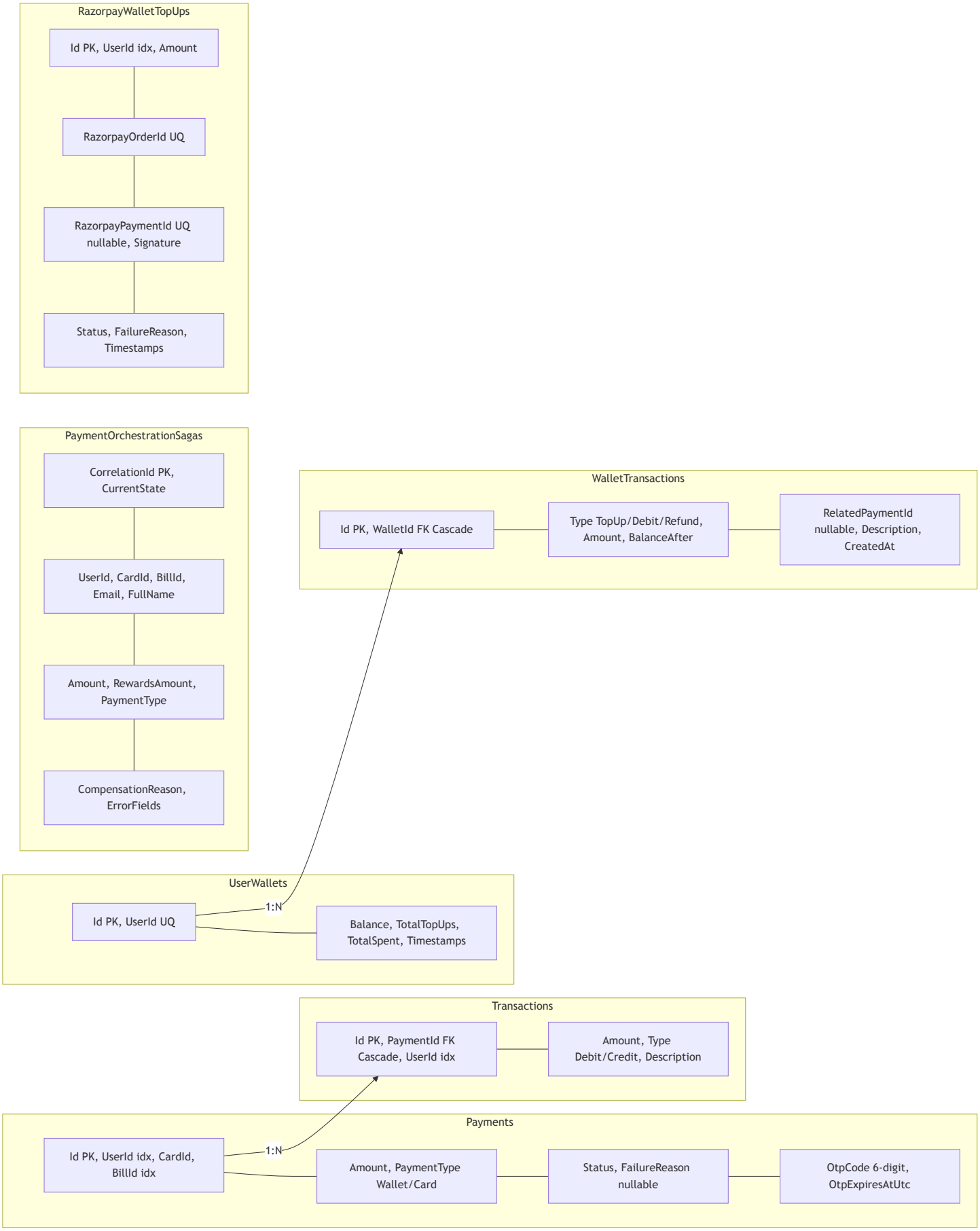
Query Filters: `IsDeleted = false` on CreditCards (global filter)

3.3 credvault_billing (6 tables)



Physical FKs: `Statements.BillId` → `Bills.Id` (Restrict), `StatementTransactions.StatementId` → `Statements.Id` (Cascade), `RewardAccounts.RewardTierId` → `RewardTiers.Id` (Restrict), `RewardTransactions.RewardAccountId` → `RewardAccounts.Id` (Cascade), `RewardTransactions.BillId` → `Bills.Id` (Restrict)

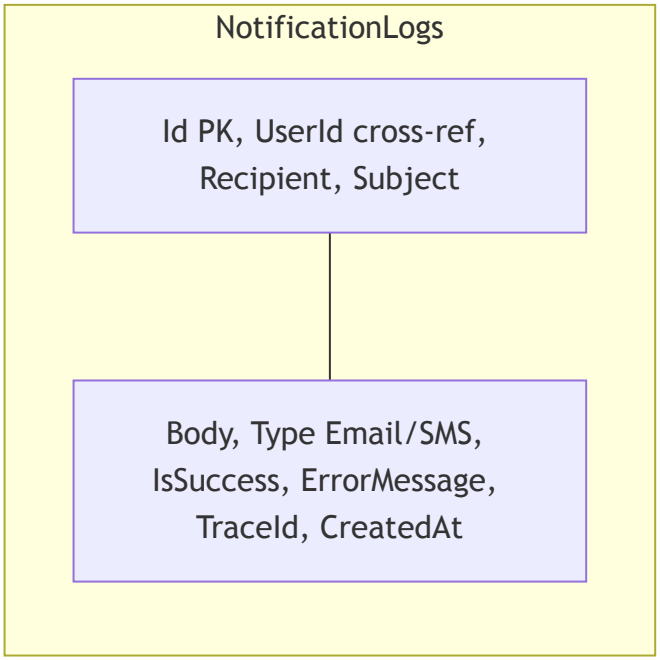
3.4 credvault_payments (6 tables)

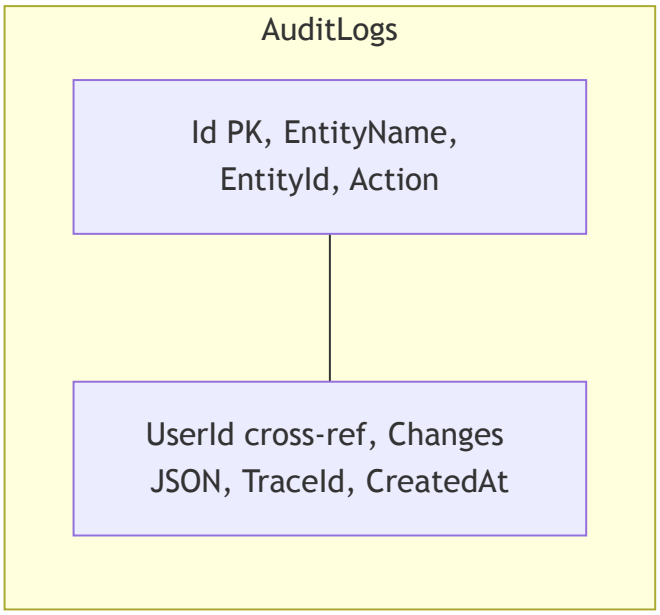


Physical FKs: `Transactions.PaymentId` → `Payments.Id` (Cascade), `WalletTransactions.WalletId` → `UserWallets.Id` (Cascade)

Notes: `PaymentOrchestrationSagas` is the MassTransit saga state table. `RazorpayOrderId` and `RazorpayPaymentId` have unique constraints for idempotency.

3.5 credvault_notifications (2 tables)





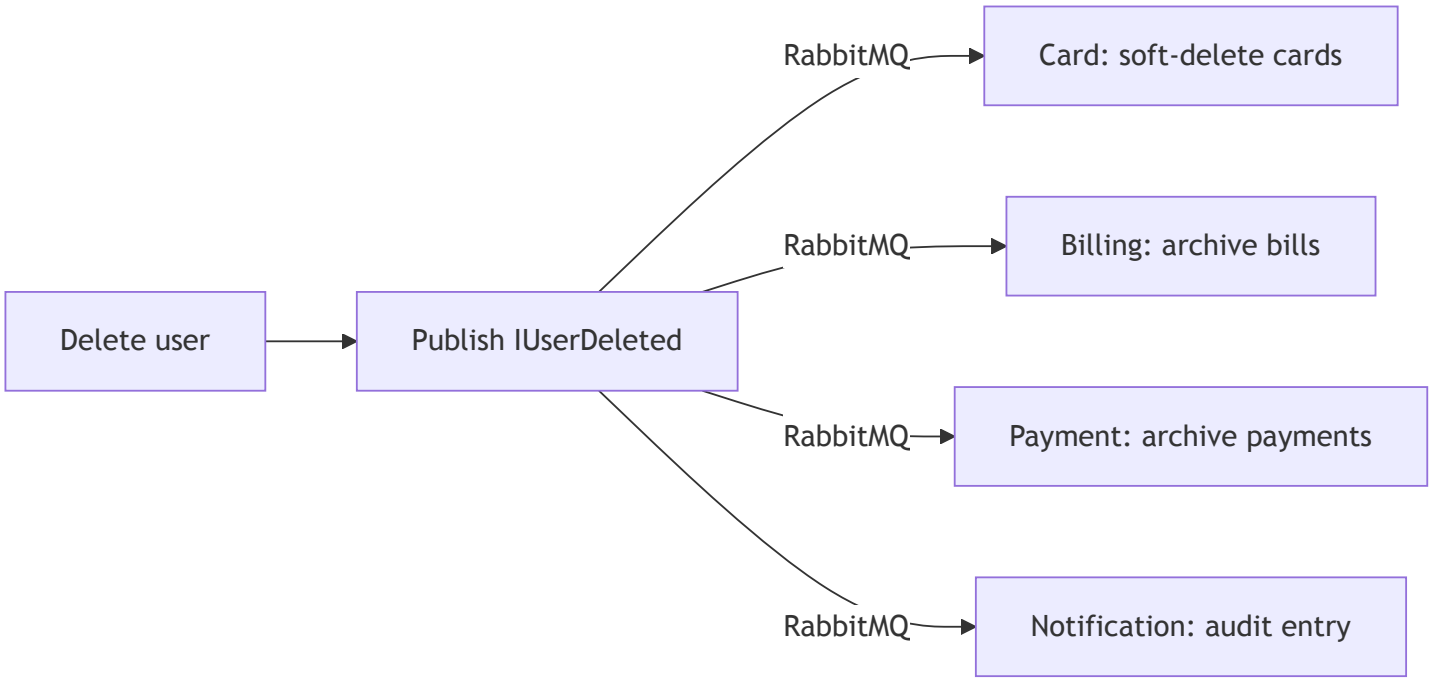
- No physical FKs in this database
- `TraceId` enables distributed request tracing
- Both tables are append-only (no updates/deletes)

4. Data Consistency Strategy

Since there are no cross-service foreign key constraints, data consistency is maintained through:

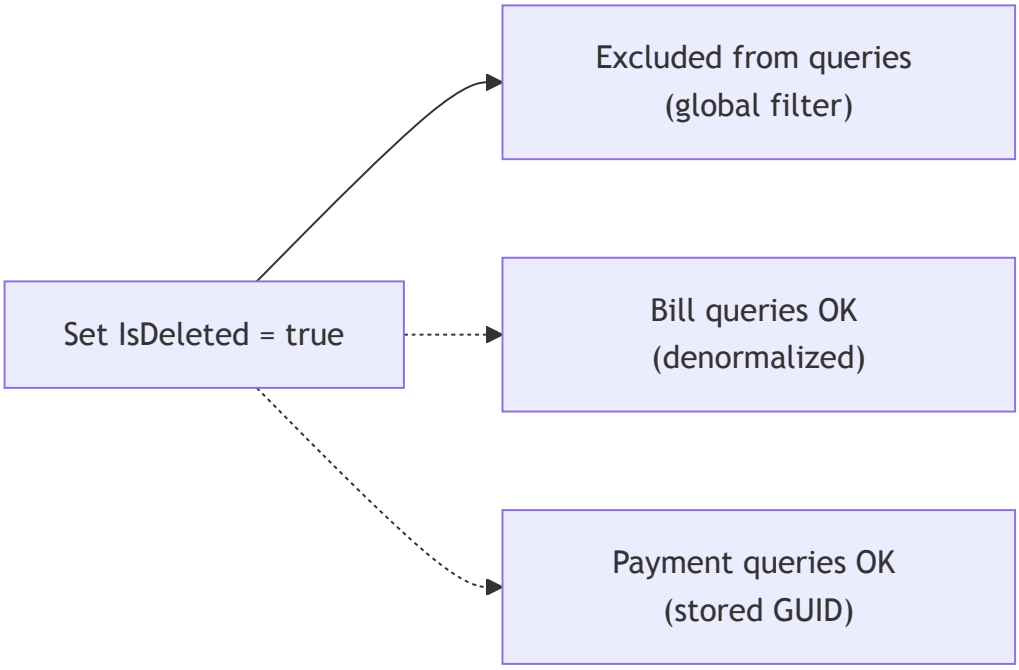
Mechanism	Purpose	Implementation
Saga Pattern	Distributed transaction consistency	MassTransit state machine with compensation
Event Publishing	Propagate changes across services	RabbitMQ pub/sub via MassTransit
Outbox Pattern	Prevent lost messages	`UseInMemoryOutbox()` on all publishers
Retry Policies	Handle transient failures	Exponential backoff: 1s → 5s → 15s
Idempotency	Safe message reprocessing	CorrelationId-based deduplication
Soft Deletes	Preserve referential integrity	`IsDeleted` flag on CreditCards
Denormalization	Reduce cross-service queries	CardLast4, CardNetwork, IssuerName stored in Bills/Statements

4.1 What Happens When a User is Deleted?



The `IUserDeleted` event consumer is implemented in each service, but the publisher in Identity Service needs to be wired up to a delete endpoint.

4.2 What Happens When a Card is Soft-Deleted?



Because Bills and Statements store CardLast4, CardNetwork, and IssuerName as denormalized columns, they remain queryable even after a card is deleted.

5. Physical Foreign Key Summary

All physical FKs exist **only within a single service's database**:

Service	Table	Column	References	Delete Behavior
Card	CreditCards	IssuerId	CardIssuers.Id	Restrict
Card	CardTransactions	CardId	CreditCards.Id	Restrict
Billing	Statements	BillId	Bills.Id	Restrict
Billing	StatementTransactions	StatementId	Statements.Id	Cascade

Service	Table	Column	References	Delete Behavior
Billing	RewardAccounts	RewardTierId	RewardTiers.Id	Restrict
Billing	RewardTransactions	RewardAccountId	RewardAccounts.Id	Cascade
Billing	RewardTransactions	BillId	Bills.Id	Restrict
Payment	Transactions	PaymentId	Payments.Id	Cascade
Payment	WalletTransactions	WalletId	UserWallets.Id	Cascade
Total physical FKs: 9	**Total cross-service FKs:** 0			

6. Index Strategy

Cross-Service Reference Indexes

Database	Table	Column	Index Type	Purpose
credvault_cards	CreditCards	UserId	Non-clustered	Find cards by user
credvault_cards	CardTransactions	UserId	Non-clustered	Find transactions by user
credvault_billing	Bills	UserId	Non-clustered	Find bills by user
credvault_billing	Bills	DueDateUtc	Non-clustered	Find overdue bills
credvault_billing	Statements	UserId	Non-clustered	Find statements by user

Database	Table	Column	Index Type	Purpose
credvault_billing	Statements	CardId	Non-clustered	Find statements by card
credvault_billing	Statements	PeriodEndUtc	Non-clustered	Find statements by period
credvault_billing	RewardAccounts	UserId	**Unique**	One reward account per user
credvault_payments	Payments	UserId	Non-clustered	Find payments by user
credvault_payments	Payments	BillId	Non-clustered	Find payments by bill
credvault_payments	UserWallets	UserId	**Unique**	One wallet per user
credvault_payments	WalletTransactions	CreatedAtUtc	Non-clustered	Find transactions by date
credvault_payments	RazorpayWalletTopUps	RazorpayOrderId	**Unique**	Idempotent webhook handling
credvault_payments	RazorpayWalletTopUps	RazorpayPaymentId	**Unique**	Idempotent webhook handling

End of Database Architecture Document